CARD 1 OF 8 THE BASIC RULE **QUESTION**

Standard **pendent or upright** sprinkler, **unobstructed** construction. How far below the ceiling does the deflector go?

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CARD 2 OF 8 UNOBSTRUCTED? **QUESTION**

What makes construction count as **unobstructed**? Give the two ways it qualifies.

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CARD 3 OF 8 OBSTRUCTED **QUESTION**

Under **obstructed** construction (solid beams or joists), what are the main ways to place a standard pendent or upright deflector?

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CARD 4 OF 8 METAL DECK **QUESTION**

Under a **corrugated metal deck**, where do you measure the deflector distance **from**?

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*FSM · Sprinkler Fact Cards — NFPA 13 (2019) Deflector Position · sheet 1 FRONT — print double-sided, **flip on LONG edge** · cut on the dashed lines*

CARD 2 OF 8 UNOBSTRUCTED? **ANSWER**

Members that **don't block heat or water** enough to hurt the sprinklers. Two ways in:

1. Open members: openings make up **at least 70%** of the member's cross-section, and the member is no deeper than the smallest opening (think open-web bar joists).

2. Wide spacing: members more than **7½ ft on center** (any construction except panel).

NFPA 13 (2019) · Ch. 3

Trap: solid wood joists 16 in. on center. That's obstructed, so the 1–12 in. rule doesn't apply as-is.

CARD 1 OF 8 THE BASIC RULE **ANSWER**

Between 1 in. and 12 in. below the ceiling.

And it has to hold that range **across the whole area the head covers**, not just at the spot where the head hangs.

Exception: concealed, recessed and flush heads go where their listing puts them.

NFPA 13 (2019) · Ch. 10

Trap: using 1–12 in. for **sidewalls**. This rule covers standard pendents and uprights only.

CARD 4 OF 8 METAL DECK **ANSWER**

Deck up to 3 in. deep: measure from the **bottom** of the deck.

Deck deeper than 3 in.: measure from the **highest point** of the deck (up inside the flute).

NFPA 13 (2019) · Ch. 9

On the job: on a deep roof deck, measuring from the low rib makes the head look higher than it really is. Measure up into the flute.

CARD 3 OF 8 OBSTRUCTED **ANSWER**

Pick one:

- **1–6 in. below the bottom** of the structural members, and no more than **22 in.** below the deck
- **At or above the bottom** of the member, up to 22 in. below the deck, if you meet the beam obstruction rules
- **In every bay**, 1–12 in. below the ceiling

There are also special cases for composite wood joists and concrete tees.

NFPA 13 (2019) · Ch. 10

Trap: forgetting the 22 in. cap. Deep beams can't push the head lower and lower.

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 INSULATION **QUESTION**

Insulation is fastened to the underside of the roof. Where do you measure the deflector distance to, if it's **flat**, and if it **sags**?

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CARD 6 OF 8 HEAT COLLECTORS **QUESTION**

A head ended up 20 in. below an unobstructed ceiling. Can you fix it by adding a small metal **heat collector** over it?

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CARD 7 OF 8 PEAKS & SLOPES **QUESTION**

Sprinklers at the **peak** of a pitched roof: how far down can the deflector be? And which way should the deflector face?

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CARD 8 OF 8 OTHER HEADS **QUESTION**

The 1–12 in. rule is for standard pendents and uprights. What range does a standard **sidewall** get? Name one other sprinkler type with its own range.

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CARD 6 OF 8 HEAT COLLECTORS **ANSWER**

No. NFPA 13 doesn't allow heat collectors to help a sprinkler activate. The head has to be moved into the right range.

Why (Annex): objects above a head can slow it down when the fire isn't directly under it. And a head far below the ceiling can't spray and cool the ceiling and structure near it.

NFPA 13 (2019) · Ch. 9 · Annex A

Trap: calling a heat collector a "fix." It's a violation, and so is the low head under it.

CARD 5 OF 8 INSULATION **ANSWER**

Flat against the roof: measure to the underside of the insulation.

Sagging: measure to **halfway** between its high and low points.

Sag over 6 in.: measure to the **high point**.

The deflector can never sit above the insulation's low point. Only batt insulation, or insulation that holds against a 3 lb/ft² uplift, counts.

NFPA 13 (2019) · Ch. 9

Trap: measuring to insulation that isn't batt and can't hold a 3 lb/ft² uplift. It doesn't count as the ceiling.

CARD 8 OF 8 OTHER HEADS **ANSWER**

Standard sidewall: deflector **4–6 in.** below the ceiling. Some horizontal sidewalls are listed for 6–12 or 12–18 in. under noncombustible or limited-combustible ceilings.

Others: upright CMSA **6–8 in.** (unobstructed). Upright ESFR K-14 **3–12 in.**

NFPA 13 (2019) · Ch. 10 · Ch. 13 · Ch. 14

Trap: one number for every head. Check which section covers the sprinkler before you set the drop.

CARD 7 OF 8 PEAKS & SLOPES **ANSWER**

Peak: no more than **36 in.** straight down from the peak. Steep roofs and saw-tooth roofs have their own rules.

Orientation: deflectors run **parallel** to the ceiling or roof slope. A head right at the peak gets a **level** deflector.

A roof pitched **2 in 12 or less** counts as flat.

NFPA 13 (2019) · Ch. 10

Trap: measuring the 36 in. along the slope. It's a vertical measurement, except under saw-tooth roofs.

sheet 2 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back